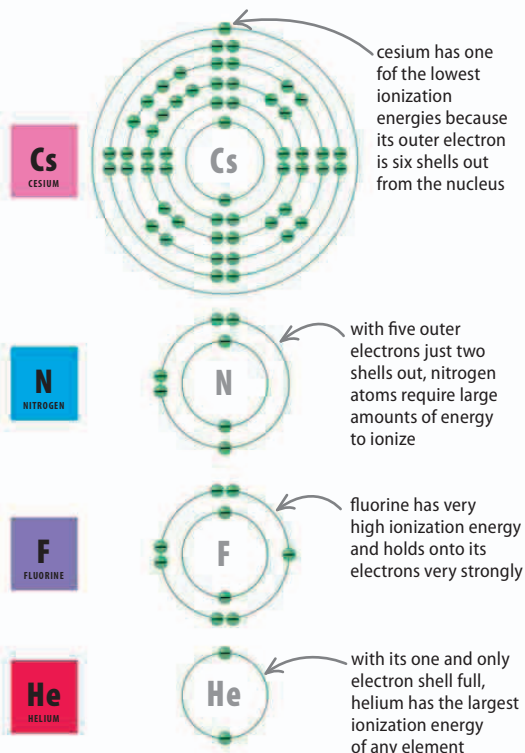


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Ionization energy is the energy needed to take an electron out of an atom, making the atom a positively charged ion. The trend in the ionization energy of elements is the reverse of that of atomic size. In the periodic table, the required energy increases left to right and decreases top to bottom. Atoms with large numbers of outer electrons require more energy to ionize by losing an electron because their shells are held more tightly, closer to the nucleus. Large atoms, with outer electrons located far from the nucleus, lose them more easily.



Trends in the table are not always followed: the atoms of **zirconium** and **hafnium** are almost identical in size, even though hafnium has 32 more electrons!

Ekasilicon

After developing the periodic table, Dmitri Mendeleev used it to predict the properties of elements that had yet to be discovered—left as gaps in the table. He described element 32 as *ekasilicon*, predicting its melting point, color, density, and chemical characteristics. In 1886, *ekasilicon*—eventually named germanium—was isolated, and matched Mendeleev's predictions.

